

- 1 A researcher wants to find out how many adults in a large town use the internet at least once a week. The researcher has formulated a suitable question to ask. For each of the following methods of taking a sample of the adults in the town, give a reason why it may be biased.

Method A: Ask people walking along a particular street between 9 am and 5 pm on one Monday.

Method B: Put the question through every letter box in the town and ask people to send back answers.

Method C: Put the question on the local council website for people to answer online.

[3]

- 2 A farmer has 200 apple trees. She is investigating the masses of the crops of apples from individual trees. She decides to select a sample of these trees and find the mass of the crop for each tree.

(a) Explain how she can select a random sample of 10 different trees from the 200 trees. [2]

The masses of the crops from the 10 trees, measured in kg, are recorded as follows.

23.5 27.4 26.2 29.0 25.1 27.4 26.2 28.3 38.1 24.9

(b) For these data find

- the mean,
- the sample standard deviation.

[2]

(c) Show that there is one outlier at the upper end of the data. How should the farmer decide whether to use this outlier in any further analysis of the data? [3]

- 3 A recruitment company advertises vacancies on their website. Information on the salaries for 36 of these vacancies is given in Fig. 7. The data have been grouped.

Salary in thousands of pounds	20 –	25 –	30 –	35 –	40 –	45 –	50 – 55
Number of vacancies	3	6	6	12	3	3	3

Fig. 7

- (a) For these salaries, calculate estimates of
- the mean,
 - the sample standard deviation.

Give your answers to the nearest pound.

[4]

- (b) Explain why your values are only estimates.

[1]

- (c) Give a reason why it would not be appropriate to use the mean calculated in part (a) as an estimate of the mean salary for all vacancies in the country.

[1]

- (d) Another vacancy has an annual salary of £52 573. This was not included in the table. Without further calculation, state how the mean salary would be affected if it were to be recalculated including this value. [1]

- 4 The managing director of an international internet communications company wishes to investigate the number of internet users and the number of mobile phone users for different countries in Eastern Europe. He wishes to identify a country with the potential to increase the number of internet users so that he can consider investing in that country.

Fig. 11.1 shows all the countries of Eastern Europe.

Country (in Eastern Europe)	Mobile phone subscribers (millions)	Internet users (millions)
Albania	3.50	1.30
Belarus	10.68	2.64
Bosnia and Herzegovina	3.35	1.42
Bulgaria	10.78	3.40
Croatia	4.97	2.23
Czech Republic	12.97	6.68
Estonia	2.07	0.97
Hungary	11.58	6.18
Kosovo	0.56	missing
Moldova	4.08	1.33
Montenegro	1.13	0.28
Poland	50.84	22.45
Romania	22.70	7.79
Serbia	9.14	4.11
Slovakia	6.10	4.06
Slovenia	2.25	1.30
Ukraine	59.34	7.77

Source: CIA World Factbook

Fig. 11.1

- (a) Are the countries in Fig. 11.1 a sample or a population? Explain your answer.

[1]

- (b) These data have been used to construct the scatter diagram in Fig. 11.2. Use your knowledge of the large data set to comment on the correlation in the scatter diagram.

[2]

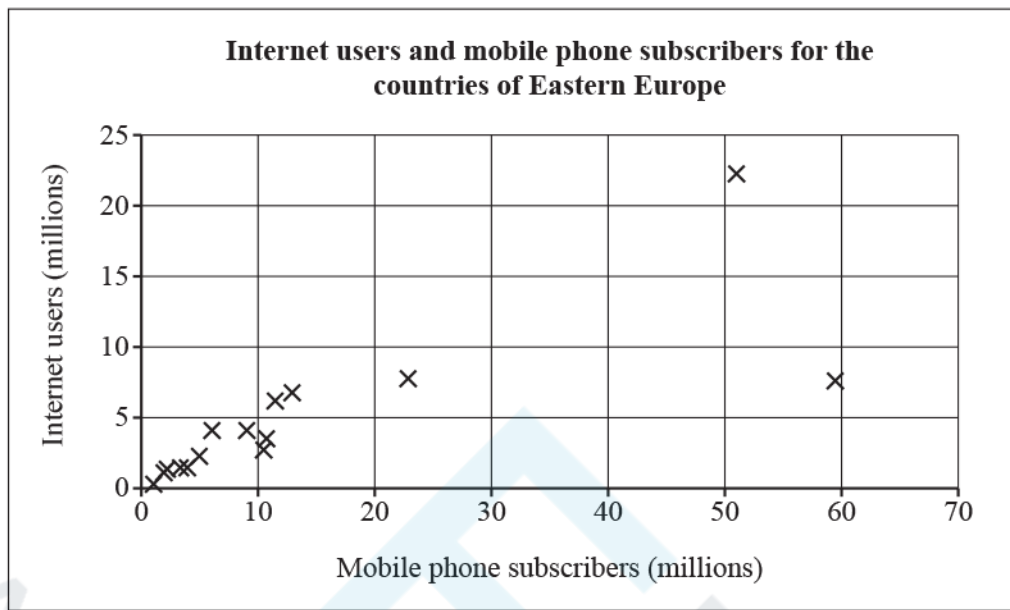


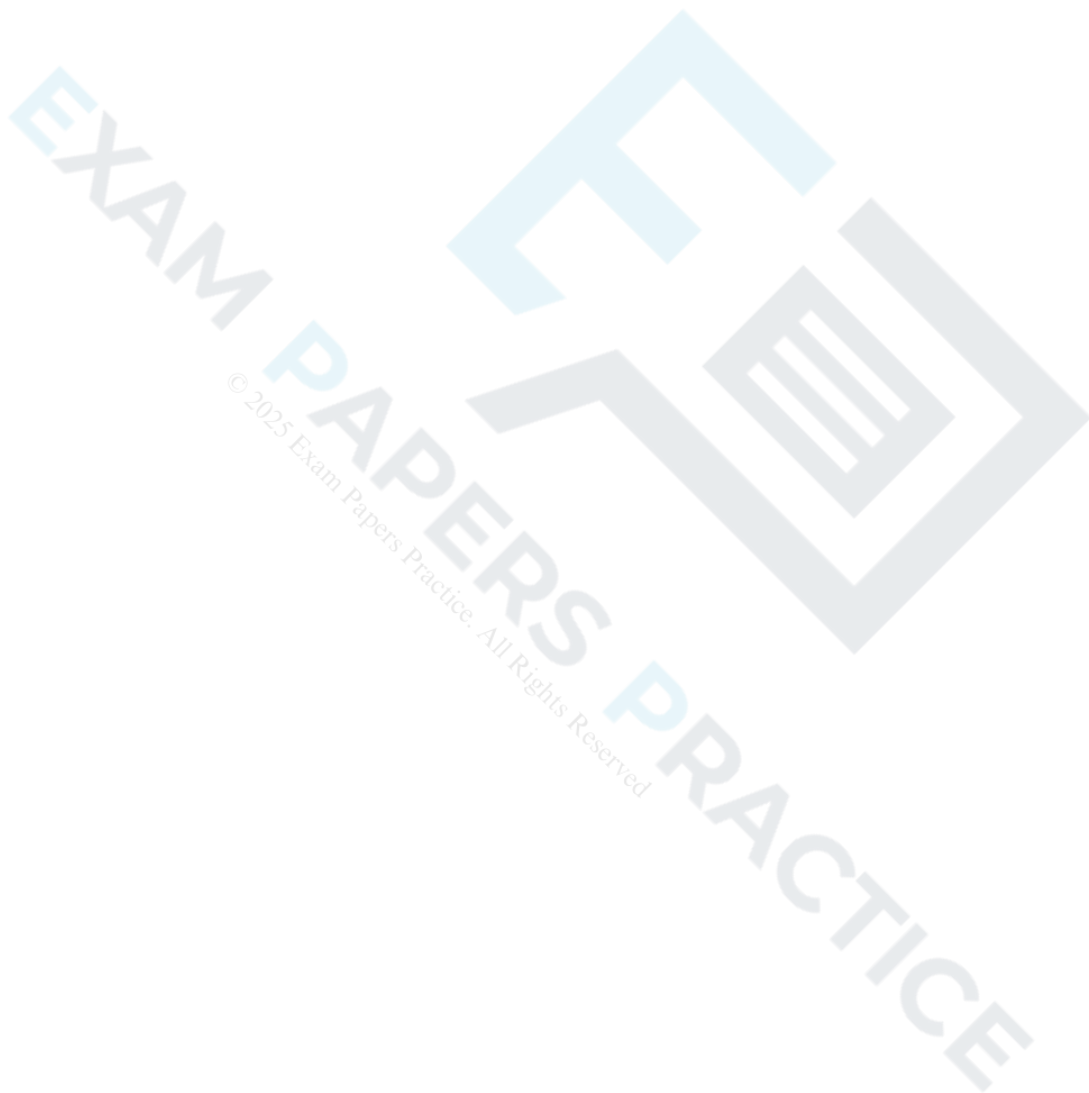
Fig. 11.2

- (c) Use the scatter diagram to identify a country that appears to have high potential to increase the number of internet users. Give a reason for your choice.

[2]

- (d) Having decided on the country he wishes to invest in, the director will select a sample of 20 marketing consultants from that country to contact for information and advice. He has found a website listing 600 marketing consultants. Give clear instructions for the director on how to select a simple random sample of 20 marketing consultants from the 600.

[3]



- 5 (a) (i) The pre-release data shows that the total population of the 239 countries in the world in 2016 was 7174654290. The populations of a sample of 10 countries are given in Fig. 9.1.

Country	Population
Tuvalu	10 782
Equatorial Guinea	722 254
Somalia	10 428 043
Denmark	5 569 077
Burma	55 746 253
Norway	5 147 792
Botswana	2 155 784
Rwanda	12 337 138
Sint Maarten	39 689
Swaziland	1 419 623

Fig. 9.1

Show that the mean population per country for the whole world is much larger than the mean population per country for this sample.

[3]

- (ii) Rebecca takes a large number of different samples of 10 countries. She finds that the mean population per country is usually smaller for the sample than it is for the whole world. Explain whether this suggests that the sampling was not random.

[2]

- (b) Fig. 9.2 shows data for Norway.

Country	Population	GDP per capita (US\$)	Health expenditure (% of GDP)
Norway	5 147 792	55 400	9.1

Fig. 9.2

Calculate Norway's health expenditure per person in US\$.

[2]

- (c) As part of an investigation into the factors which might be associated with life expectancy, the scatter diagrams in Fig. 9.3 are drawn. The line of best fit and the corresponding value of the square of the correlation coefficient are also shown for each scatter diagram.

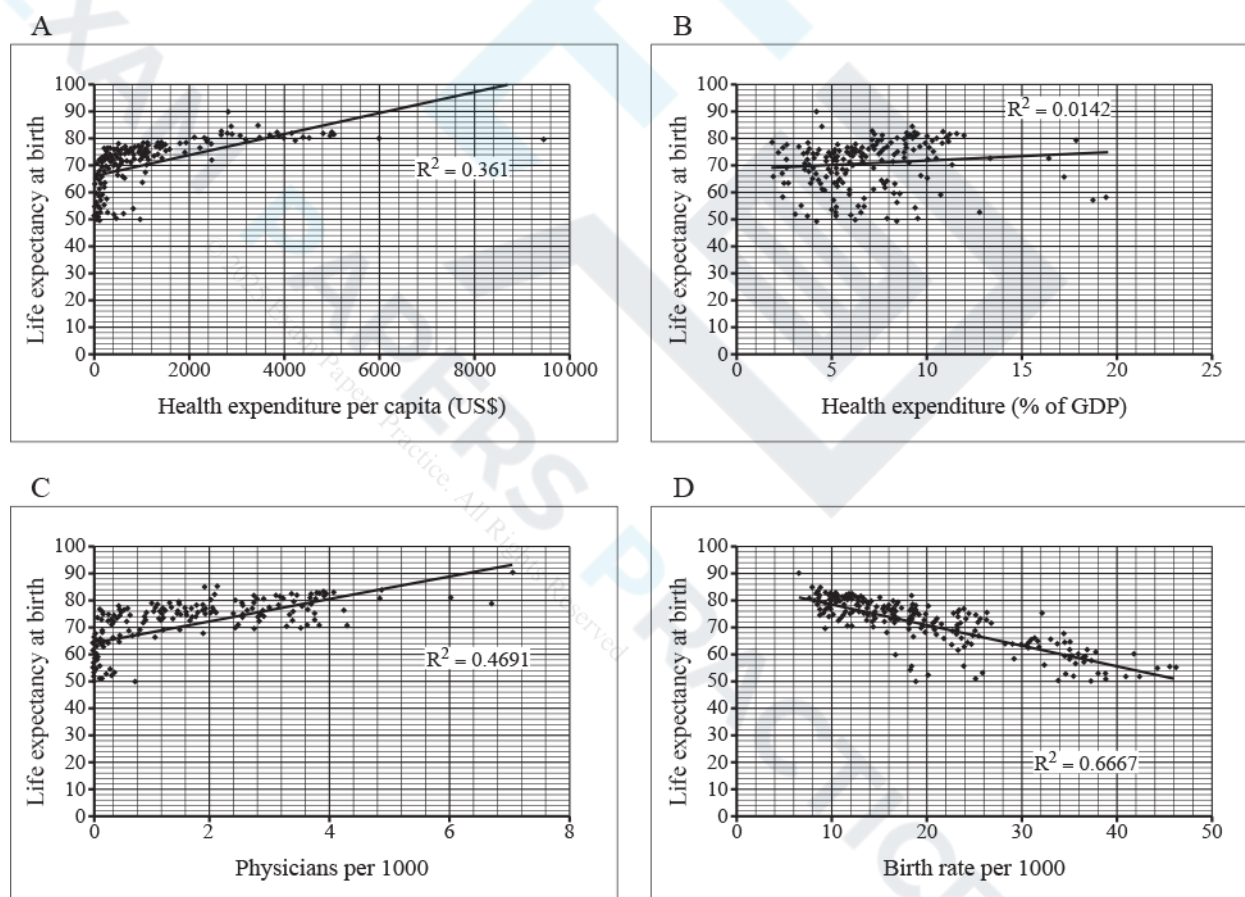


Fig. 9.3

- (i) Which of the four factors appears to have the strongest positive association with life expectancy at

birth? Give a reason for your answer.

[2]

- (ii) Explain why the line of best fit in scatter diagram B is not a good model for the relationship between the two variables.

[1]

- 6 In a certain school there are 273 boys and 327 girls. A proportional stratified sample of 50 pupils is required.

How many girls should be in the sample?

[2]

- 7 The pre-release data set consists of a subset (see <http://www.ocr.org.uk/Images/308749-units-h630-and-h640-large-data-set-lds-sample-assessment-material.xls>) of the data about countries from the CIA World Facebook.
- Data for Croatia shows that the average number of mobile phone subscribers per 1000 population is 1111.72.

(a) Explain how this average can be greater than 1000.

[1]

It is assumed that, once the data have been cleaned, the average number of phones per 1000 population in different countries can reasonably be approximated by a Normal distribution with mean 996 and standard deviation 407.

(b) State two ways in which the data may have been cleaned.

[2]

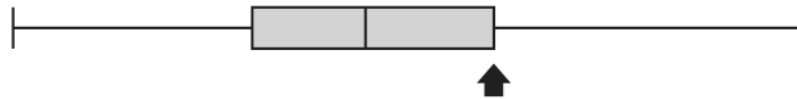


Fig. 11.1

Fig. 11.1 shows a box plot of the cleaned data.

(c) Give the name of the measure which the arrow is pointing to.

[2]

(d) Use the approximate Normal distribution to estimate a value for this measure.

[2]

Fatima thinks that the mean number of mobile phone subscribers per 1000 population for the countries of the world has increased since the data were collected. In order to test this he obtains some up-to-date data for a random sample of countries and uses software to conduct the appropriate hypothesis test at the 5% level of significance. The output from the software is shown in Fig. 11.2.

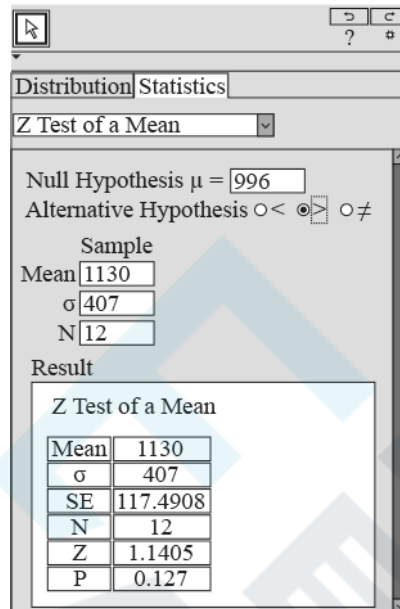


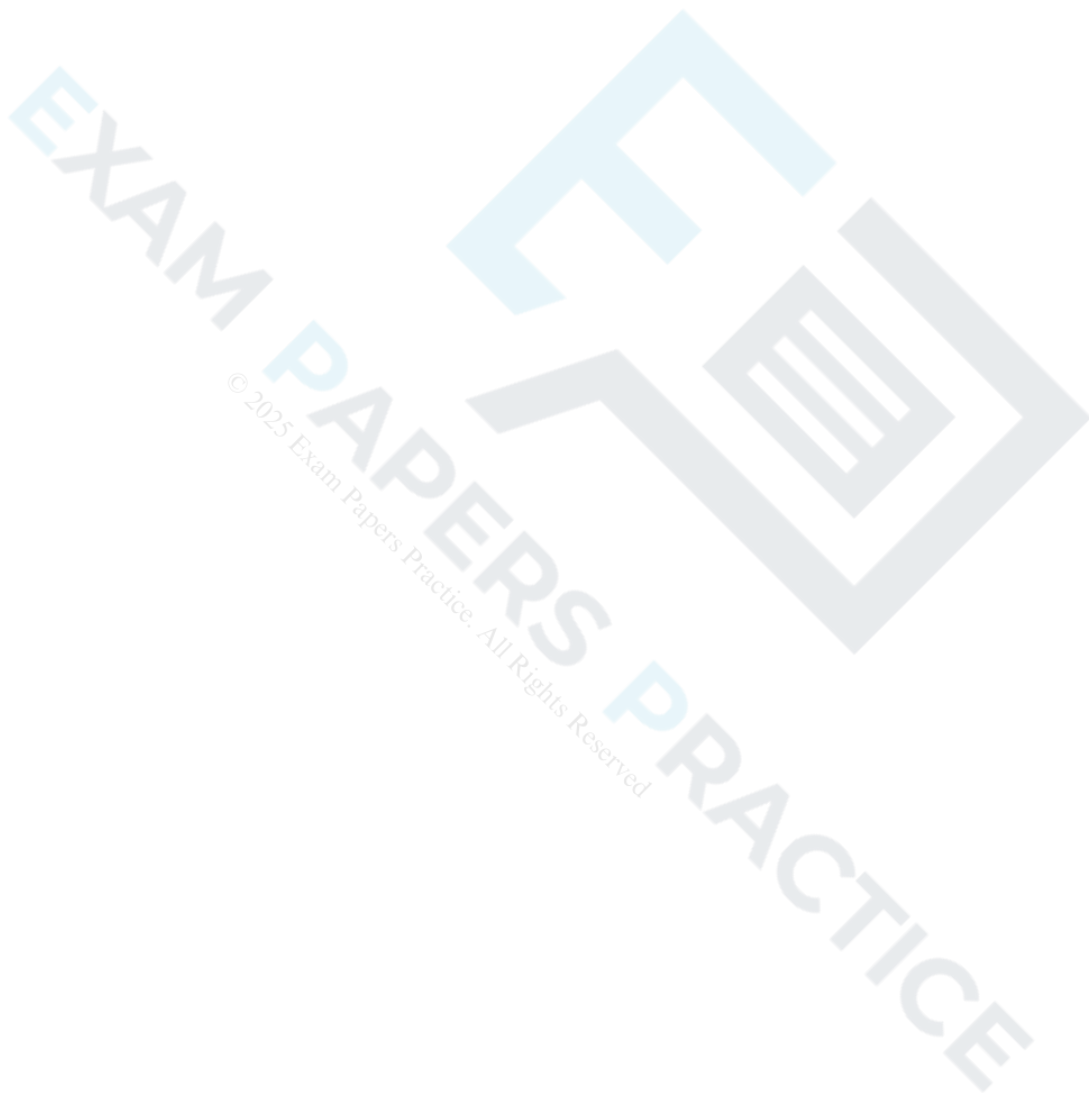
Fig. 11.2

(e) State the conclusion of Fatima's test, explaining your reasoning.

[4]

(f) With reference to the pre-release data set, comment on the sample size Fatima has chosen.

[1]



- 8 The pre-release material contains data concerning the death rate per thousand people and the birth rate per thousand people in all the countries of the world. The diagram in Fig. 11.1 was generated using a spreadsheet and summarises the birth rates for all the countries in Africa.

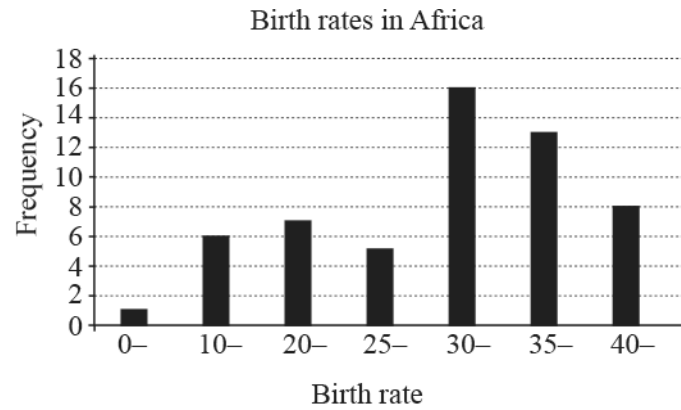


Fig. 11.1

- (a) Identify **two** respects in which the presentation of the data is incorrect.

[2]

Fig. 11.2 shows a scatter diagram of death rate, y , against birth rate, x , for a sample of 55 countries, all of which are in Africa. A line of best fit has also been drawn.

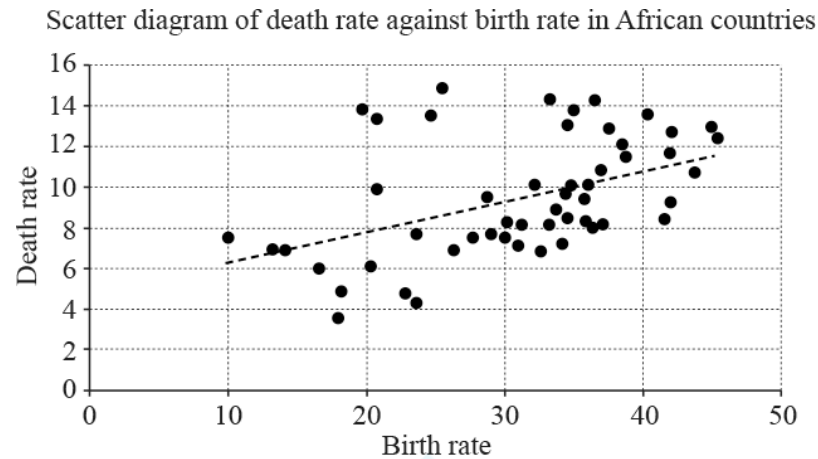


Fig. 11.2

The equation of the line of best fit is $y = 0.15x + 4.72$.

(b) (i) What does the diagram suggest about the relationship between death rate and birth rate?

[1]

(ii) The birth rate in Togo is recorded as 34.13 per thousand, but the data on death rate has been lost. Use the equation of the line of best fit to estimate the death rate in Togo.

[1]

- (iii) Explain why it would not be sensible to use the equation of the line of best fit to estimate the death rate in a country where the birth rate is 5.5 per thousand. [1]

- (iv) Explain why it would not be sensible to use the equation of the line of best fit to estimate the death rate in a Caribbean country where the birth rate is known. [1]

(v) Explain why it is unlikely that the sample is random.

[1]

Including Togo there were 56 items available for selection.

(c) Describe how a sample of size 14 from this data could be generated for further analysis using systematic sampling.

[2]

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- 9 The pre-release material includes data on unemployment rates in different countries. A sample from this material has been taken. All the countries in the sample are in Europe. The data have been grouped and are shown in Fig 14.1.

Unemployment rate	0–	5–	10–	15–	20–	35–50
Frequency	15	21	5	5	2	2

Fig. 14.1

A cumulative frequency curve has been generated for the sample data using a spreadsheet. This is shown in Fig. 14.2.

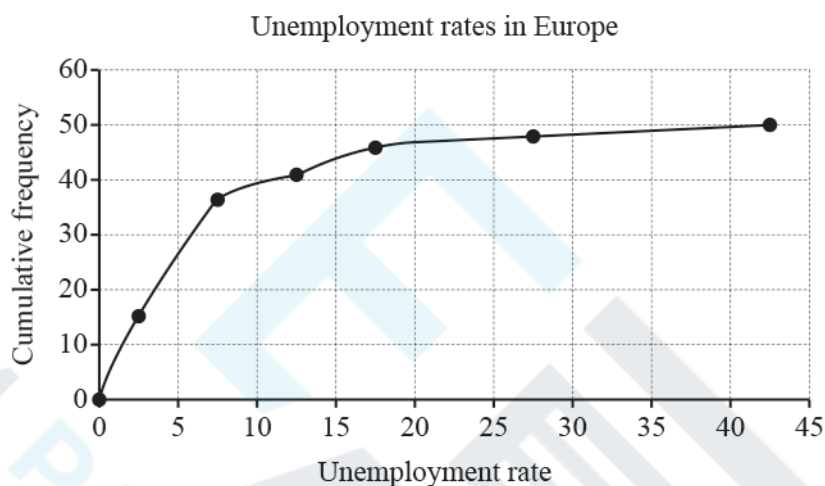


Fig. 14.2

Hodge used Fig. 14.2 to estimate the median unemployment rate in Europe. He obtained the answer 5.0. The correct value for this sample is 6.9.

- (a) (i) There is a systematic error in the diagram.

- Identify this error.
- State how this error affects Hodge's estimate.

[2]

(ii) There is another factor which has affected Hodge's estimate.

- Identify this factor.
- State how this factor affects Hodge's estimate.

[2]

(b) Use your knowledge of the pre-release material to give another reason why any estimation of the median unemployment rate in Europe may be unreliable.

[1]

(c) Use your knowledge of the pre-release material to explain why it is very unlikely that the sample has been randomly selected from the pre-release material.

[1]

The scatter diagram shown in Fig. 14.3 shows the unemployment rate and life expectancy at birth for the 47 countries in the sample for which this information is available.

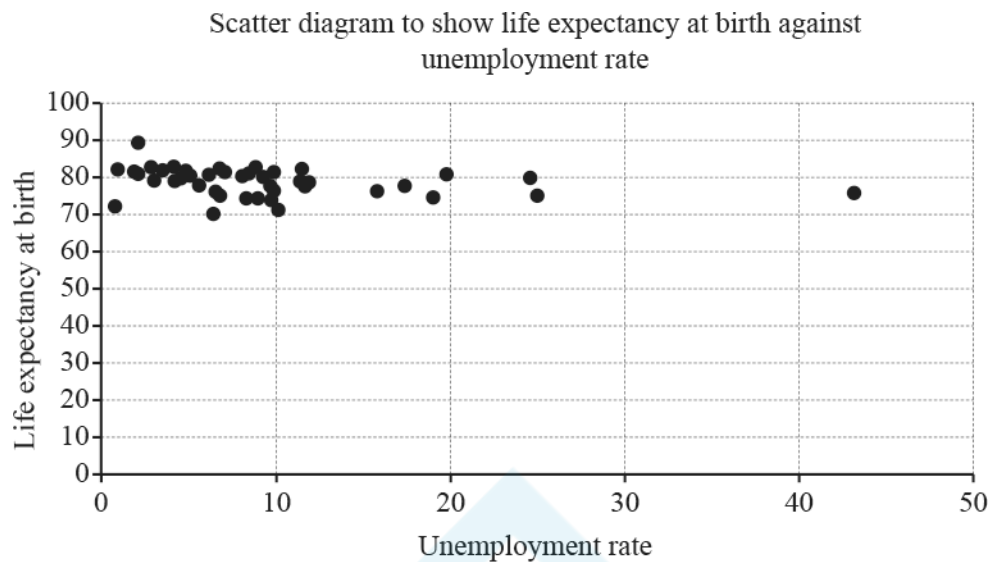


Fig. 14.3

The product moment correlation coefficient for the 47 items in the sample is -0.2607 .

The p -value associated with $r = -0.2607$ and $n = 47$ is 0.0383.

- (d) Does this information suggest that there is an association between unemployment rate and life expectancy at birth in countries in Europe? [2]

Hodge uses the spreadsheet tools to obtain the equation of a line of best fit for this data.

- (e) The unemployment rate in Kosovo is 35.3, but there is no data available on life expectancy. Is it reasonable to use Hodge's line of best fit to estimate life expectancy at birth in Kosovo? [1]

- 10 A class representative is investigating whether pupils at his school believe that school meals are satisfactory. The class representative has an alphabetical list of all 619 pupils in the school saved on a spreadsheet. He decides to select a sample of 50 pupils, and considers two different sampling procedures.

Procedure A.

Assign a distinct random number to each pupil. Select the 50 pupils with the smallest random numbers.

Procedure B.

Generate a 2-digit random number. Use this random number to select a starting point on the list according to the rule shown in Fig. 8. Select the pupil identified by this rule, and then select every 12th pupil on the list after this, stopping when a sample of 50 has been obtained.

Random number	00–10	11–20	21–30	31–40	41–50	51–70	71–80	81–99
Starting point	1	2	3	4	5	6	7	8

Fig. 8

- (a) Explain why procedure A will generate a simple random sample.

[1]

(b) Identify **two** features of procedure **B** that prevent it from generating a simple random sample.

[2]

(c) Describe how you could generate a random sample of size 50 from the 619 pupils using systematic sampling.

[1]

- 11 The spreadsheet output in Fig. 12.1 gives some information about all the countries that won more than 10 gold medals in the London 2012 Olympic Games.

A	B	C	D	E	F	G	H
1	Country	population	GDP per capita (US\$)	Total GDP (US\$)	Gold medals	Silver medals	Bronze medals
2	United States	318 892 103	52 800	1.68×10^{13}	46	29	29
3	China	1 355 692 576	9800	1.33×10^{13}	38	27	22
4	United Kingdom	63 742 977	37 300	2.38×10^{12}	29	17	19
5	Russia	142 470 272	18 100	2.58×10^{12}	24	25	33
6	Korea, South	49 039 986	33 200	1.63×10^{12}	13	8	7
7	France	66 259 012	35 700	2.37×10^{12}	11	11	12
8	Germany	80 996 685	39 500	3.20×10^{12}	11	19	14

Fig. 12.1

- (a) Give a spreadsheet formula for calculating the value in cell D2 using other cell values in Fig. 12.1. [1]

The statistics in Fig. 12.2 are for all the countries in the pre-release data.

	Population	GDP per capita (US\$)	Total GDP (US\$)
Lower Quartile	4.587×10^5	4525	2.09×10^9
Median	5.623×10^6	13750	7.73×10^{10}
Upper Quartile	2.116×10^7	31750	6.72×10^{10}

Fig. 12.2

- (b) Explain whether or not the statements below are consistent with the information given in Fig. 12.1 and Fig 12.2.

Statement A

Countries with larger populations are more likely to win Olympic gold medals.

Statement B

Countries with larger total GDP are more likely to win Olympic gold medals.

[3]

There were approximately 10 500 Olympic competitors in 2012. The population of the world was approximately 7 000 000 000 in 2012.

A geography student assumed that Olympic competitors are randomly and uniformly scattered across the population of the world. The student calculated that the population of a country with two Olympic competitors would be approximately 1 300 000.

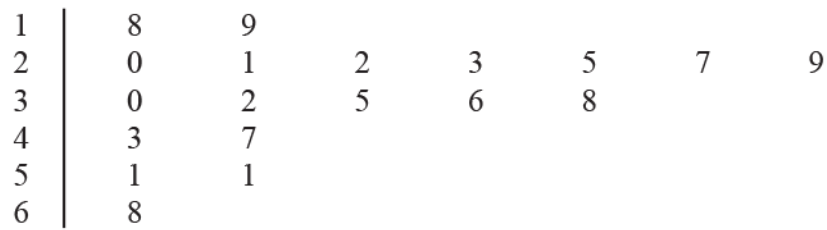
- (c) Use your knowledge of the large data set to explain whether the geography student's assumption is realistic.

[1]

- 12 Qasim has just opened a café in Burnton. He decides to conduct some market research on his customers.

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One Monday morning at 11 am he asked every customer in the café to fill out a questionnaire which included asking for the customer's age. The results he collected are shown in the stem-and-leaf diagram in Fig. 7.1.



Key 2 | 0 represents an age of 20.

Fig. 7.1

- (a) What name is given to the sampling method used by Qasim? [1]

- (b) Explain why the sample does not represent a simple random sample of all the customers who use Qasim's café. [1]

- (c) Describe the shape of the distribution of the ages in the sample. [1]

- (d) For the data in Fig. 7.1 find
- the median,
 - the interquartile range.

[3]

Kai works at Qasim's café. He believes that Qasim's sample data may not be representative of all Qasim's customers. One week he asks every customer to fill in the questionnaire. Summary statistics for the customers who filled in the questionnaire are shown in Fig. 7.2.

Number of respondents	237
Age of youngest person	7
Lower quartile	27
Median	31
Upper quartile	39
Age of oldest person	81

Fig. 7.2

- (e) Comment on whether the statistics in Fig. 7.2 provide any evidence to support Kai's belief.

[2]

- 13(a) The pre-release material includes data concerning crude death rates in different countries of the world. Fig. 14.1 shows some information concerning crude death rates in countries in Europe and in Africa.

	Europe	Africa
n	48	56
minimum	6.28	3.58
lower quartile	8.50	7.31
median	9.53	8.71
upper quartile	11.41	11.93
maximum	14.46	14.89

Fig. 14.1

Use your knowledge of the large data set to suggest a reason why the statistics in Fig. 14.1 refer to only 48 of the 51 European countries.

[1]

- (b) Use the information in Fig. 14.1 to show that there are no outliers in either data set.

[3]

- (c) The crude death rate in Libya is recorded as 3.58 and the population of Libya is recorded as 6 411 776.

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Calculate an estimate of the number of deaths in Libya in a year.

[1]

- (d) The median age in Germany is 46.5 and the crude death rate is 11.42. The median age in Cyprus is 36.1 and the crude death rate is 6.62.

Explain why a country like Germany, with a higher median age than Cyprus, might also be expected to have a higher crude death rate than Cyprus.

[1]

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- (e) Fig. 14.2 shows a scatter diagram of median age against crude death rate for countries in Africa and Fig. 14.3 shows a scatter diagram of median age against crude death rate for countries in Europe.

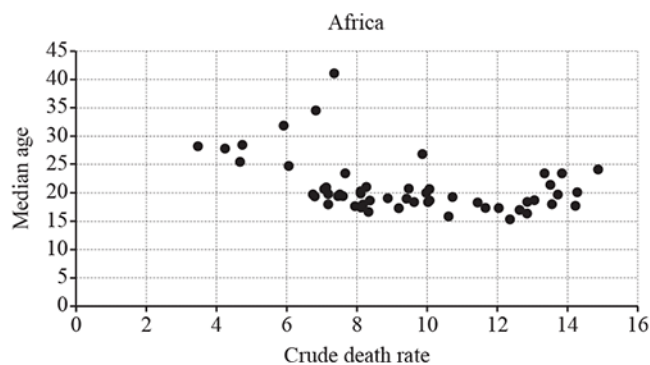


Fig. 14.2

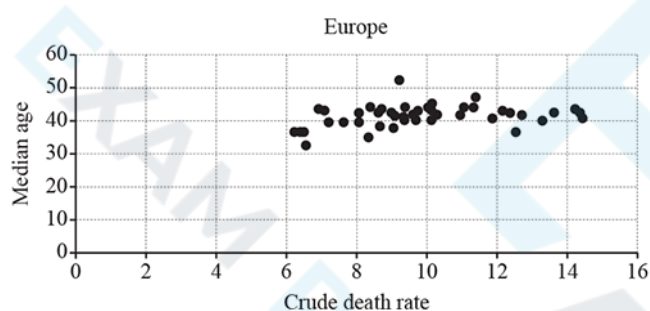


Fig. 14.3

The rank correlation coefficient for the data shown in Fig. 14.2 is -0.281206.

The rank correlation coefficient for the data shown in Fig. 14.3 is 0.335 215.

Compare and contrast what may be inferred about the relationship between median age and crude death rate in countries in Africa and in countries in Europe. [2]

14(a) A researcher is conducting an investigation into the number of portions of fruit adults consume each day. The researcher decides to ask 50 men and 50 women to complete a simple questionnaire.

State the type of sampling procedure the researcher is using.

[1]

(b) Write down one disadvantage of this sampling procedure.

[1]

(c) The researcher represents the data in Fig. 3.1.

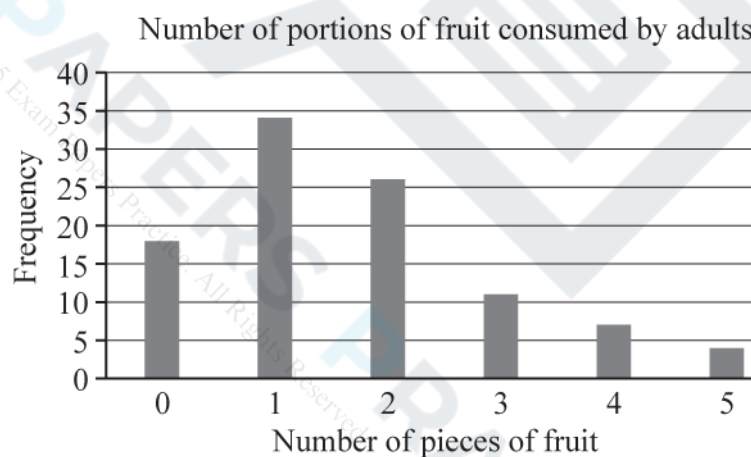


Fig. 3.1

Describe the shape of the distribution.

[1]

(d) The data are summarised in the frequency table in Fig. 3.2.

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Number of portions of fruit	0	1	2	3	4	5
Number of adults	18	34	26	11	7	4

Fig. 3.2

For the data in Fig. 3.2, use your calculator to find

- the mean,
- the standard deviation.

Give your answers correct to 2 decimal places.

[2]

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- (e) A second researcher chooses a proportional stratified sample of 100 children from years 5 and 6 in a certain primary school. There are 220 children to choose from. In year 5 there are 125 children, of whom 81 are boys.

How many year 5 girls should be included in the sample?

[1]

- (f) The second researcher found that the mean number of portions of fruit consumed per day by the children in this sample was 1.61 and the standard deviation was 0.53.

Comment on the amount of fruit consumed per day by the children compared to the amount of fruit consumed per day by the adults.

[2]

15(a) Fig. 10.1 shows a sample collected from the large data set.

BMI is defined as $\frac{\text{mass of person in kilograms}}{\text{square of person's height in metres}^2}$

Sex	Age in years	Mass in kg	Height in cm	BMI
Male	38	77.6	164.8	28.57
Male	17	63.5	170.3	21.89
Male	18	68.0	172.3	22.91
Male	18	57.2	172.2	19.29
Male	19	77.6	191.2	21.23
Male	24	72.7	177.0	23.21
Male	25	92.5	177.9	29.23
Male	26	70.4	159.4	27.71
Male	31	77.5	174.0	25.60
Male	34	132.4	182.2	39.88
Male	38	115.0	186.4	33.10
Male	40	112.1	171.7	38.02

Fig. 10.1

Calculate the mass in kg of a person with a BMI of 23.56 and a height of 181.6 cm, giving your answer correct to 1 decimal place.

[2]

- (b) Fig. 10.2 shows a scatter diagram of BMI against age for the data in the table. A line of best fit has also been drawn.

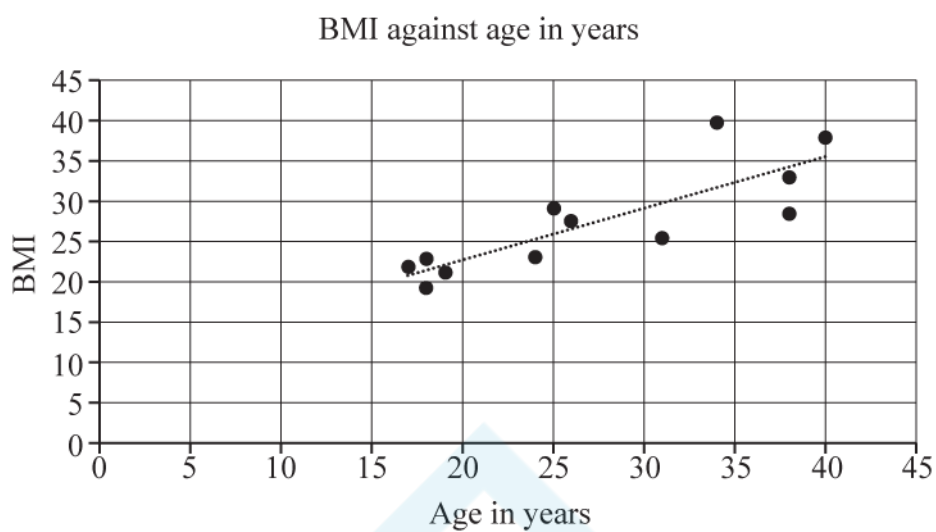


Fig. 10.2

Describe the correlation between age and BMI.

[1]

(c) Use the line of best fit to estimate the BMI of a 30-year-old man.

[1]

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(d) Explain why it would not be sensible to use the line of best fit to estimate the BMI of a 60-year-old man.

[1]

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- (e) Use your knowledge of the large data set to suggest two reasons why the sample data in the table may not be representative of the population.

- (f) Once the data in the large data set had been cleaned there were 196 values available for selection. Describe how a sample of size 12 could be generated using systematic sampling so that each of the 196 values could be selected in the sample.

16(a) Rosella is carrying out an investigation into the age at which adults retire from work in the city where she lives. She collects a sample of size 50, ensuring this comprises of 25 randomly selected retired men and 25 randomly selected retired women.

State the name of the sampling method she uses.

[1]



(b) Fig. 8.1 shows the data she obtains in a frequency table and Fig. 8.2 shows these data displayed in a histogram.

Age in years at retirement	45 –	50 –	55 –	60 –	65 –	70 –	75 – 80
Frequency Density	0.4	1.8	2.4	2.2	1.8	1.2	0.2

Fig. 8.1

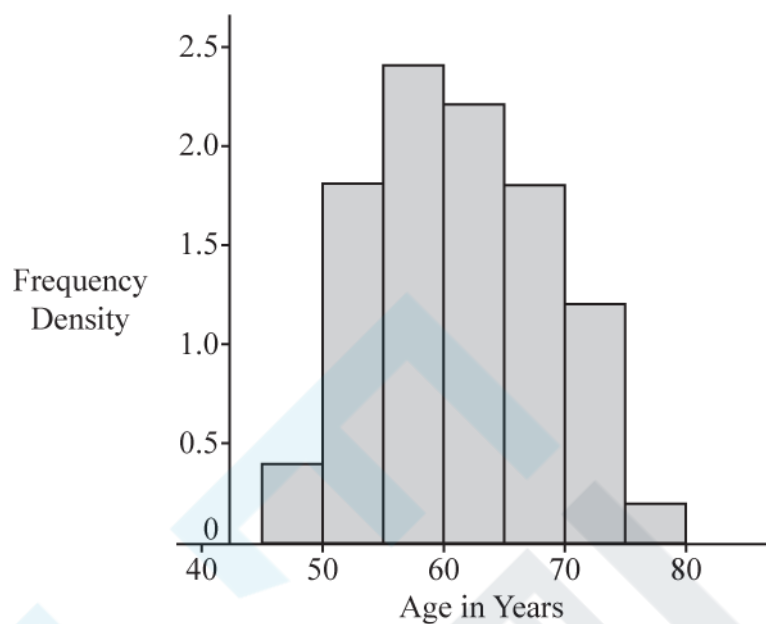


Fig. 8.2

How many people in the sample are aged between 50 and 55?

[1]

- (c) Rosella obtains a list of the names of all 4960 people who have retired in the city during the previous month.

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Describe how Rosella could collect a sample of size 200 from her list using

- systematic sampling such that every item on the list could be selected,
- simple random sampling.

[4]



- (d) Rosella collects two simple random samples, one of size 200 and one of size 500, from her list. The histograms in Fig. 8.3 show the data from the sample of size 200 on the left and the data from the sample of 500 on the right.

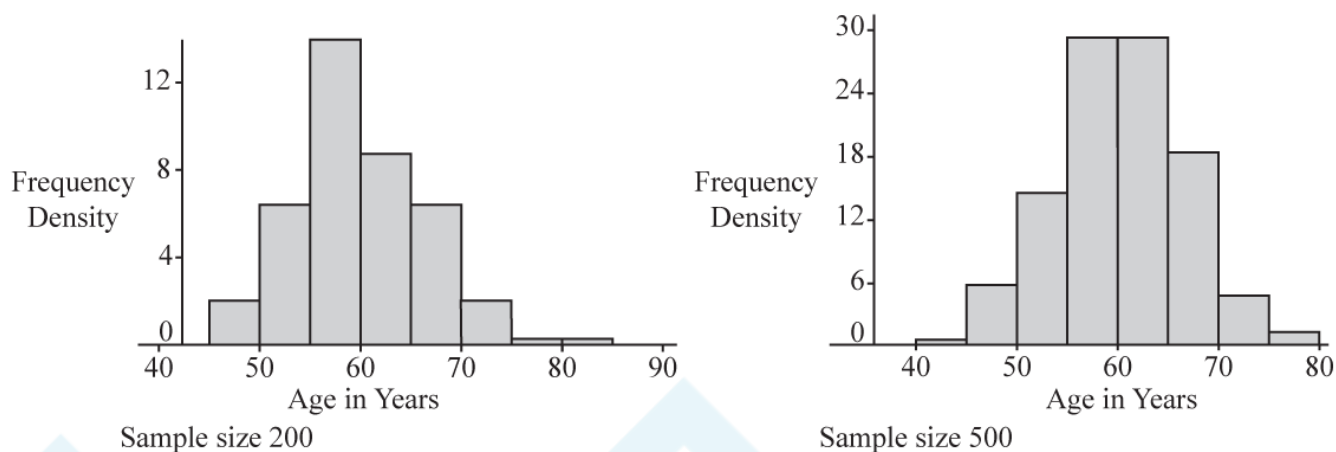


Fig. 8.3

With reference to the histograms shown in Fig. 8.2 and Fig. 8.3, explain why it appears reasonable to model the age of retirement in this city using the Normal distribution. [1]

- (e) Summary statistics for the sample of 500 are shown in Fig. 8.4.

Statistics	
n	500
Mean	60.0515
σ	6.5717
s	6.5783
Σx	30025.7601
Σx^2	1824686.322
Min	36.0793
Q1	55.2573
Median	59.9202
Q3	64.4239
Max	81.742

Fig. 8.4

Use an appropriate Normal model based on the information in Fig. 8.4 to estimate the number of people aged over 65 who retired in the city in the previous month.

[4]

- (f) Identify a limitation in using this model to predict the number of people aged over 65 retiring in the following month.

[1]

- 17 A company supplies computers to businesses. In the past the company has found that computers are kept by businesses for a mean time of 5 years before being replaced. Claud, the manager of the company, thinks that the mean time before replacing computers is now different.

Describe how Claud could obtain a cluster sample of 120 computers used by businesses the company supplies.

[1]

18(a) The pre-release material contains information concerning median house prices over the period 2004 – 2015. A spreadsheet has been used to generate a time series graph for two areas: the London borough of “Barking and Dagenham” and “North West”. This is shown together with the raw data in Fig. 11.1.

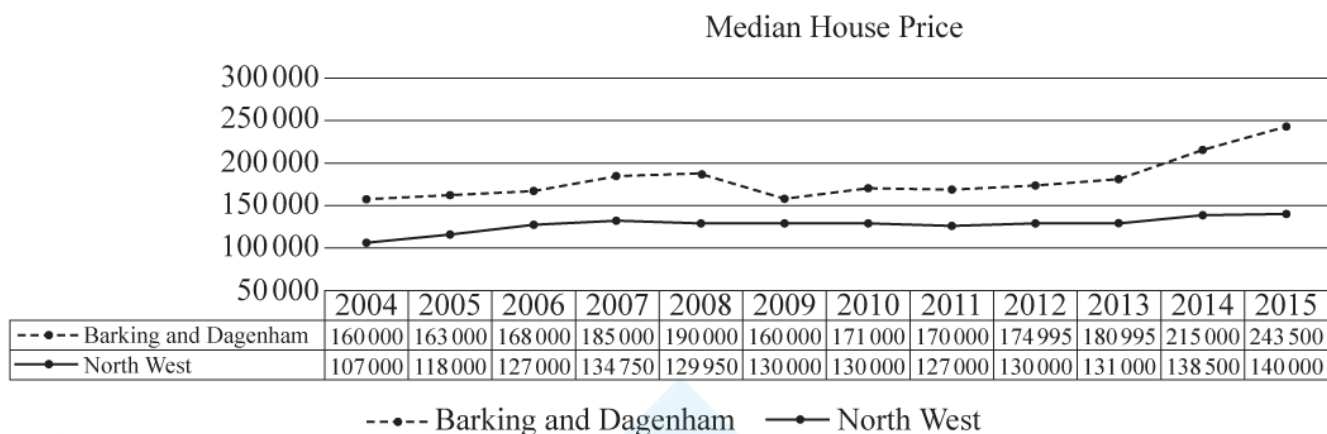


Fig. 11.1

Dr Procter suggests that it is unusual for median house prices in a London borough to be consistently higher than those in other parts of the country.

Use your knowledge of the large data set to comment on Dr Procter's suggestion.

[1]

- (b) Dr Procter wishes to predict the median house price in Barking and Dagenham in 2016. She uses the spreadsheet function LINEST to find the equation of the line of best fit for the given data. She obtains the equation

$P = 4897Y - 9\,657\,847$, where P is the median house price in pounds and Y is the calendar year, for example 2015.

Use Dr Procter's equation to predict the median house price in Barking and Dagenham in

- 2016
- 2017.

[2]

- (c) Professor Jackson uses a simpler model by using the data from 2014 and 2015 only to form a straight-line model.

Find the equation Professor Jackson uses in her model.

[2]

- (d) Use Professor Jackson's equation to predict the median house price in Barking and Dagenham in

- 2016
- 2017.

[2]

- (e) Professor Jackson carries out some research online. She finds some information about median house prices in Barking and Dagenham, which is shown in Fig. 11.2.

2016	2017
£290 000	£300 000

Fig. 11.2

Comment on how well

- Dr Procter's model fits the data,
- Professor Jackson's model fits the data.

[2]

- (f) Explain which, if any, of the models is likely to be more reliable for predicting median house prices in Barking and Dagenham in 2020.

[1]

19(a) The pre-release material contains information concerning median house prices, recycling rates and employment rates. Fig. 13.1 shows a scatter diagram of recycling rate against employment rate for a random sample of 33 regions.

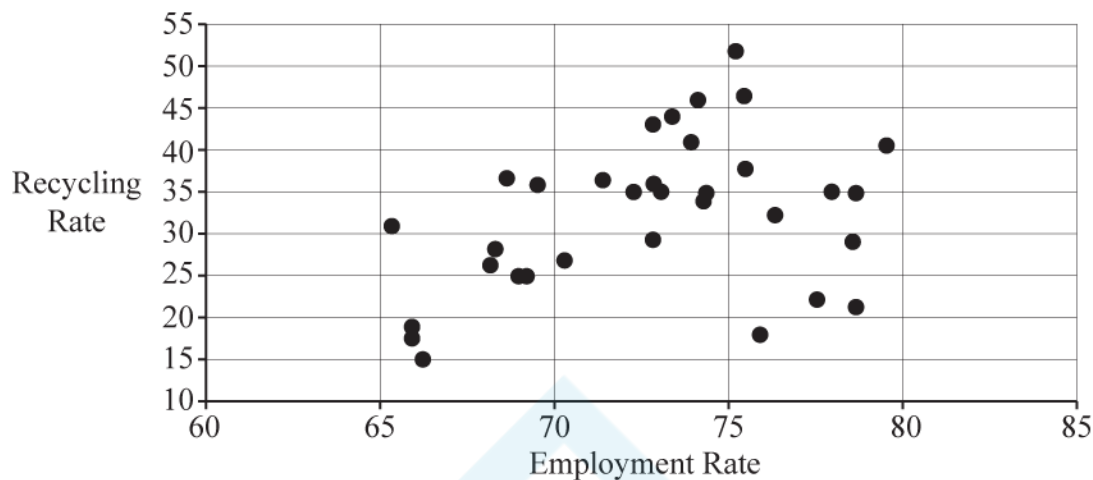


Fig. 13.1

The product moment correlation coefficient for this sample is 0.37154 and the associated p -value is 0.033.

Lee conducts a hypothesis test at the 5% level to test whether there is any evidence to suggest there is positive correlation between recycling rate and employment rate. He concludes that there is no evidence to suggest positive correlation because $0.033 \approx 0$ and $0.37154 > 0.05$.

Explain whether Lee's reasoning is correct.

[2]

- (b) Fig. 13.2 shows a scatter diagram of recycling rate against median house price for a random sample of 33 regions.

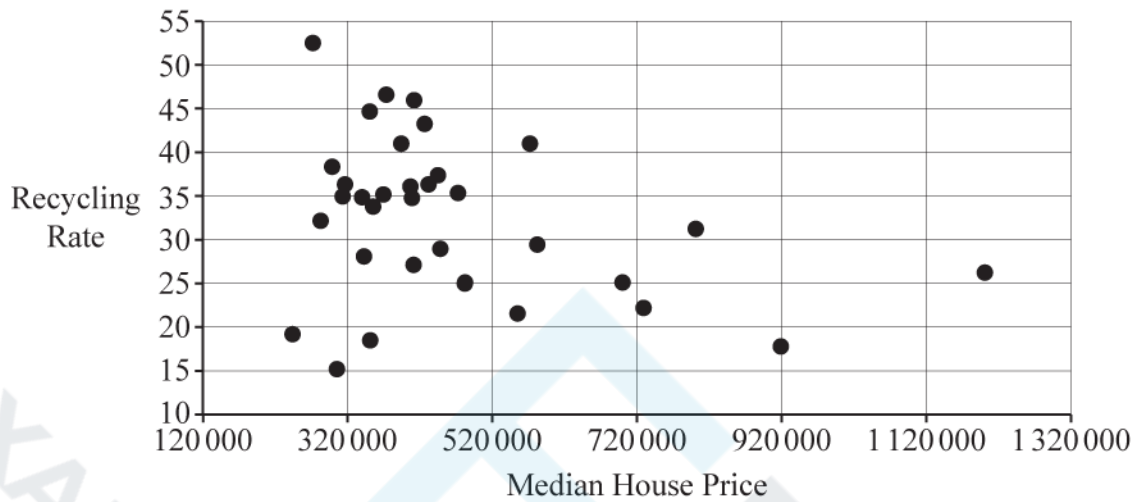


Fig. 13.2

The product moment correlation coefficient for this sample is -0.33278 and the associated p -value is 0.058 .

Fig. 13.3 shows summary statistics for the median house prices for the data in this sample.

Statistics	
n	33
Mean	465467.9697
σ	201236.1345
s	204356.2606
Σx	15360443
Σx^2	8486161617387
Min	243500
Q1	342500
Median	410000
Q3	521000
Max	1200000

Fig. 13.3

Use the information in Fig. 13.3 and Fig. 13.2 to show that there are at least two outliers.

[2]

(c) Describe the effect of removing the outliers on

- the product moment correlation coefficient between recycling rate and median house price,
- the p -value associated with this correlation coefficient, in each case explaining your answer.

[2]

(d) All 33 items in the sample are areas in London. A student suggests that it is very unlikely that only areas in London would be selected in a random sample.

Use your knowledge of the pre-release material to explain whether you think the student's suggestion is reasonable.

[1]

END OF QUESTION PAPER